Fake Profile Detection on Social Media Using Machine Learning

A PROJECT REPORT

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Problem Statement

The widespread use of social media platforms such as Twitter, Facebook, and Instagram has significantly increased communication and information sharing. However, the growing number of fake profiles has become a major concern, as these accounts are often used for spamming, phishing, identity theft, financial fraud, misinformation, and cyberbullying. Such malicious activities reduce user trust and compromise the security of online communities.

Traditional fake profile detection methods mainly rely on manual verification or rule-based approaches, which are time-consuming and ineffective against rapidly evolving fake account behaviors. These methods often fail to identify sophisticated fake profiles that closely resemble genuine users.

This project proposes a **Machine Learning-based Fake Profile Detection System** using the **Random Forest algorithm** to automatically classify social media accounts as **fake** or **genuine**. The system analyzes user profile attributes such as the number of followers, following, posts, account age, profile description, verification status, and engagement-related features to identify suspicious accounts. By leveraging machine learning, the proposed solution aims to provide accurate, fast, and scalable detection of fake profiles, thereby improving the security, reliability, and trustworthiness of social media platforms.

INTRODUCTION

Social media has become one of the most important platforms for communication, networking, business promotion, education, and entertainment. Applications such as Twitter (X), Facebook, Instagram, and LinkedIn allow millions of users to connect, share information, and interact with people worldwide. While these platforms provide numerous benefits, they also face significant security challenges due to the increasing number of fake profiles. Fake accounts are created by individuals or automated bots to spread misinformation, conduct phishing attacks, promote scams, manipulate public opinion, send spam, or steal personal information. The presence of such accounts reduces user trust and threatens the safety and reliability of online communities.

A fake profile is an account that misrepresents the identity of a person or organization. These accounts often use stolen photographs, false personal information, or automated behavior to appear genuine. Fake profiles are commonly used for identity theft, financial fraud, cyberbullying, political propaganda, fake product promotions, and malicious advertising campaigns. As social media usage continues to grow, detecting these fake accounts has become an essential task for maintaining a secure digital environment.

Traditional methods for detecting fake profiles mainly depend on manual verification or predefined rules. However, these approaches are time-consuming, difficult to maintain, and unable to adapt to constantly changing patterns of fraudulent behavior. Machine Learning (ML) provides an effective alternative by learning patterns from historical data and automatically identifying suspicious accounts. ML algorithms can analyze multiple profile features simultaneously, discover hidden relationships among them, and make accurate predictions without relying solely on manually defined rules.

Among the various machine learning algorithms, **Random Forest** is widely used because of its high classification accuracy, robustness, and ability to handle large datasets with multiple features. Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions to produce a final classification. It is less prone to overfitting than a single decision tree, performs well with both numerical and categorical data, and provides reliable predictions even when some features contain noise or missing values. These characteristics make it highly suitable for fake profile detection.

In this project, the **Twitter Fake & Real User Dataset** is used to train and evaluate the Random Forest classifier. Profile attributes such as followers count, friends count, account age, verification status, profile description, profile image information, and activity-related features are analyzed to distinguish genuine users from fake ones. The proposed system aims to classify user profiles accurately and efficiently while reducing false predictions.

The primary objectives of this project are to collect and preprocess social media profile data, extract meaningful features, train a Random Forest model, evaluate its performance using standard metrics, and automatically classify social media accounts as **fake** or **genuine**. The proposed system contributes to improving the security, reliability, and trustworthiness of social media platforms by assisting in the early detection of malicious accounts.

LITERATURE SURVEY

The rapid growth of social media platforms has transformed the way people communicate, share information, and interact online. Platforms such as Twitter (X), Facebook, Instagram, and LinkedIn have become integral parts of daily life. However, this growth has also led to a significant increase in fake profiles, bot accounts, and malicious users that exploit these platforms for identity theft, phishing, misinformation, financial fraud, spam campaigns, and cyberbullying. As a result, detecting fake profiles has become an important research area in cybersecurity and machine learning. Researchers have proposed various techniques ranging from rule-based methods to advanced machine learning and deep learning models to improve the accuracy of fake profile detection.

Early fake profile detection methods primarily relied on manually defined rules and profile verification techniques. These approaches analyzed basic account characteristics such as profile completeness, account age, number of followers, and posting frequency to identify suspicious users. Although these methods were simple to implement, they lacked adaptability and failed to detect sophisticated fake accounts that mimicked genuine user behavior. Their dependence on handcrafted rules also made them ineffective against evolving attack patterns.

To overcome these limitations, **Mahesh et al. (2024)** proposed a machine learning-based fake profile detection system using Twitter user datasets. Their study evaluated classification algorithms including Random Forest and Decision Tree by analyzing user profile features such as followers, following, posts, and account activity. The Random Forest model achieved higher accuracy and better classification performance than other algorithms due to its ensemble learning capability. However, the study focused only on Twitter data and did not evaluate the model on other social media platforms.

Further improvements were introduced by **Aditya et al. (2023)** through a heterogeneous social media analysis framework for fake profile identification. Their research employed deep learning techniques to analyze profile metadata, behavioral characteristics, and user interactions. By integrating multiple feature types, the proposed framework significantly improved detection accuracy. Nevertheless, the model required high computational resources and a large amount of labeled training data, making deployment in real-time applications more challenging.

Sansonetti et al. (2020) proposed a deep learning framework for detecting unreliable users in social media environments. Their model utilized user behavior, posting patterns, and profile attributes to distinguish genuine accounts from fake ones. Experimental results demonstrated higher accuracy than conventional machine learning methods. However, the framework relied heavily on large datasets and required extensive training, limiting its applicability to resource-constrained environments.

Another significant contribution was presented by **Van Der Walt et al. (2018)**, who investigated the use of machine learning algorithms for distinguishing fake identities from genuine users. Their research compared classifiers such as Random Forest and Support Vector Machine (SVM), concluding that Random Forest produced superior classification accuracy while remaining computationally efficient. However, the proposed model primarily analyzed static profile attributes and did not consider evolving user behavior over time.

Similarly, **Kumar et al. (2019)** developed a spammer detection framework using Random Forest and behavioral analysis for identifying fake users on social networks. The model successfully detected spam accounts with high precision and recall by analyzing user activity patterns and account information. Despite its effectiveness, the model struggled to identify newly created fake accounts that frequently changed their behavioral characteristics to evade detection.

Recent research has expanded fake profile detection beyond individual platforms. **Goyal et al. (2024)** proposed a machine learning framework for detecting fake Instagram profiles using profile metadata, follower statistics, engagement features, and account activity. Their study demonstrated that combining multiple profile attributes significantly improved classification performance. However, the approach involved extensive feature engineering and preprocessing, increasing computational complexity.

Survey-based research by **Ramalingam and Valliyammai (2018)** reviewed various machine learning, data mining, and hybrid approaches for fake profile detection across online social networks. Their work highlighted major research challenges, including dataset imbalance, feature selection, cross-platform adaptability, and scalability. The authors concluded that ensemble learning methods, particularly Random Forest, offer a reliable balance between accuracy, robustness, and computational efficiency.

Overall, existing research demonstrates that machine learning algorithms have significantly improved fake profile detection compared to traditional rule-based methods. Among these algorithms, **Random Forest** has emerged as one of the most effective classifiers due to its high accuracy, robustness against overfitting, and ability to process large feature sets. However, challenges such as cross-platform generalization, real-time detection, evolving fake account behavior, and imbalanced datasets remain active research areas. The proposed project addresses these challenges by implementing a Random Forest-based fake profile detection system using the Twitter Fake & Real User Dataset, providing an accurate, scalable, and interpretable solution for classifying social media accounts as **fake** or **genuine**.

RESEARCH GAP

Although several researchers have proposed machine learning and deep learning techniques for fake profile detection, several challenges still remain. The major research gaps identified from the literature are as follows:

1. **Platform-Specific Models:** Most existing studies focus only on a single social media platform, such as Twitter or Instagram, which limits the generalization of their models to other social networking platforms.

2. **High Computational Complexity:** Many recent approaches rely on deep learning models that require powerful hardware, large training datasets, and long training times, making them difficult to deploy in real-time applications.

3. **Limited Feature Utilization:** Several research works consider only basic profile information, ignoring important behavioral and activity-based features that can improve fake profile detection accuracy.

4. **Dataset Imbalance Issues:** Many publicly available datasets contain an unequal number of fake and genuine profiles, leading to biased models and reduced classification performance.

5. **Lack of Model Interpretability:** Deep learning models often behave as black-box systems, making it difficult to understand the reasons behind classification decisions and reducing user trust.

6. **Difficulty in Detecting Evolving Fake Profiles:** Fake account creators continuously modify profile attributes and behavior to bypass existing detection systems, reducing the effectiveness of traditional models.

7. **Need for an Efficient and Scalable Solution:** There is a requirement for a detection system that provides high accuracy, faster prediction, lower computational cost, and better scalability. Therefore, this project proposes a **Random Forest-based Fake Profile Detection System** that efficiently classifies social media accounts as **fake** or **genuine** using profile metadata and activity-related features.

PROPOSED SOLUTION WITH BLOCK DIAGRAM

To overcome the limitations of traditional fake profile detection methods that rely on manual verification and fixed rule-based techniques, this project proposes a **Machine Learning-based Fake Profile Detection System** using the **Random Forest algorithm**. The proposed system automatically classifies social media accounts as **fake** or **genuine** by analyzing various user profile attributes and behavioral features. Unlike conventional approaches, the model learns directly from profile data, enabling it to identify hidden patterns and improve detection accuracy. Random Forest is selected because of its high classification performance, robustness against overfitting, and ability to handle large datasets with multiple features.

Architecture Overview

The proposed system follows a sequential pipeline consisting of **dataset collection, data preprocessing, feature extraction, model training, prediction, and performance evaluation**. Each stage plays an important role in improving the accuracy and reliability of fake profile detection.

Steps of the Proposed Solution

1. Dataset Collection

The proposed system uses the **Twitter Fake & Real User Dataset** obtained from Kaggle. The dataset consists of two separate files containing fake user profiles and genuine user profiles. These datasets are merged to create a single dataset for model development. Each profile contains various attributes such as username, followers count, friends count, tweets count, verification status, account age, description, and other profile-related information. The combined dataset is then divided into training and testing sets for model development and evaluation.

2. Data Preprocessing

To improve data quality and ensure effective model training, several preprocessing operations are performed.

- Merge fake and genuine user datasets.
- Remove duplicate records.
- Handle missing or null values.
- Encode categorical attributes into numerical values.
- Clean unnecessary information.
- Normalize the dataset wherever required.

These preprocessing steps help reduce noise, improve consistency, and prepare the data for machine learning.

3. Feature Extraction

After preprocessing, important profile attributes are selected for classification. The selected features include:

- Followers Count
- Friends (Following) Count
- Tweets/Posts Count
- Verified Status
- Account Age
- Description
- Other profile-related attributes

Selecting the most relevant features improves the prediction accuracy while reducing unnecessary computations.

4. Random Forest Classification

The extracted features are provided to the **Random Forest classifier**. The dataset is divided into **80% training data and 20% testing data**. During training, Random Forest constructs multiple decision trees using different subsets of the training data. Each tree independently predicts whether a profile is fake or genuine. The final prediction is determined using the **majority voting** technique, making the model more robust and reducing the possibility of overfitting.

5. Prediction

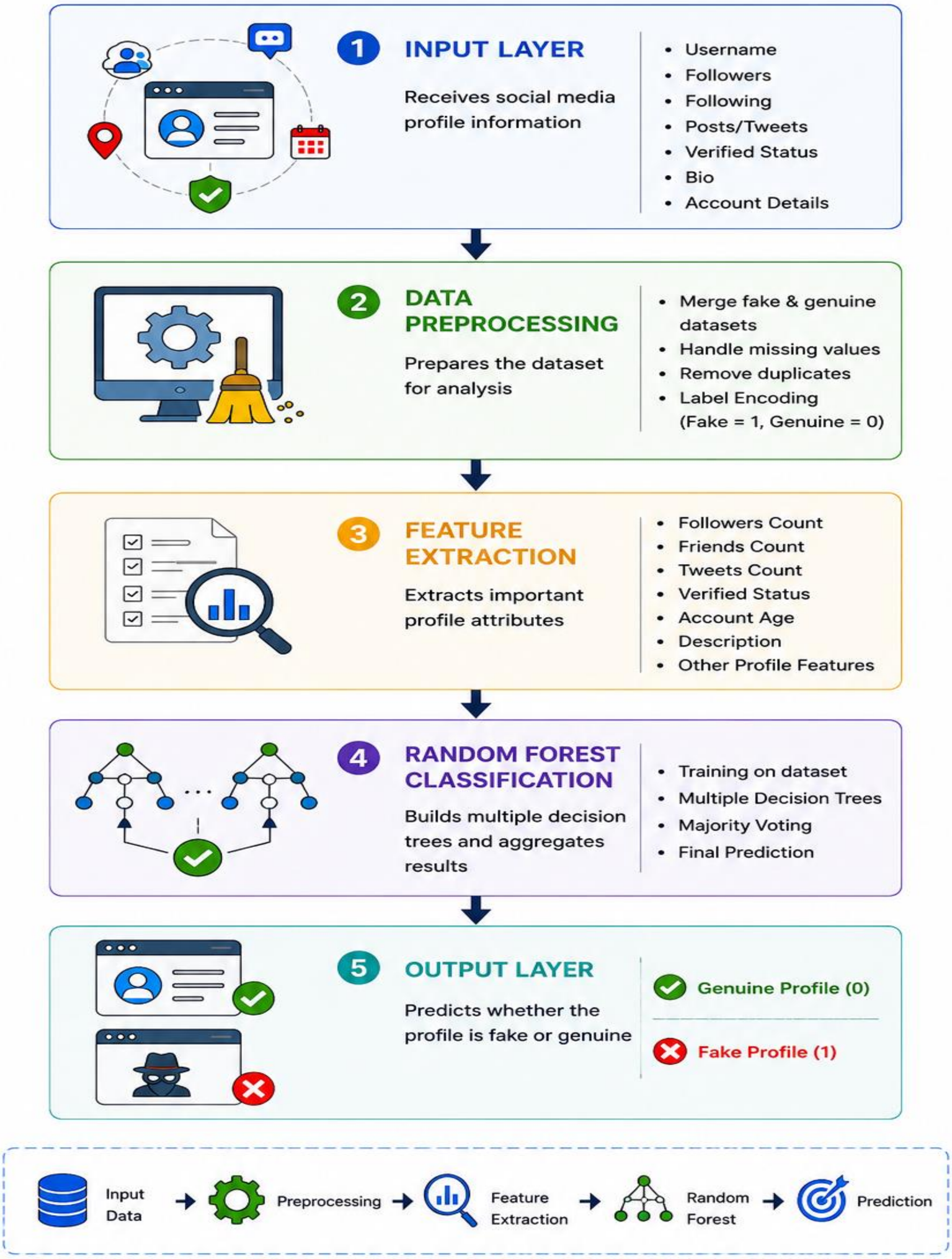
Once the model is trained, it receives a new social media profile as input. Based on the learned patterns, the Random Forest classifier predicts whether the account is **Fake Profile (1)** or **Genuine Profile (0)**. The prediction is generated quickly and accurately, enabling automatic identification of suspicious accounts.

6. Evaluation and Performance Analysis

The performance of the proposed fake profile detection system is evaluated using standard classification metrics, including:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

The experimental results demonstrate that the Random Forest classifier provides high classification accuracy while maintaining low computational complexity. Compared to traditional rule-based detection methods, the proposed system offers a more reliable, scalable, and efficient solution for identifying fake social media profiles.


1 INPUT LAYER
Receives social media profile information
• Username
• Followers
• Following
• Posts/Tweets
• Verified Status
• Bio
• Account Details
2 DATA PREPROCESSING
Prepares the dataset for analysis
• Merge fake & genuine datasets
• Handle missing values
• Remove duplicates
• Label Encoding (Fake = 1, Genuine = 0)
3 FEATURE EXTRACTION
Extracts important profile attributes
• Followers Count
• Friends Count
• Tweets Count
• Verified Status
• Account Age
• Description
• Other Profile Features
4 RANDOM FOREST CLASSIFICATION
Builds multiple decision trees and aggregates results
• Training on dataset
• Multiple Decision Trees
• Majority Voting
• Final Prediction
5 OUTPUT LAYER
Predicts whether the profile is fake or genuine
Genuine Profile (0)
Fake Profile (1)
Input Data
Preprocessing
Feature Extraction
Random Forest
Prediction

RESULT AND DISCUSSION

The proposed **Fake Profile Detection System** was developed using the **Random Forest** machine learning algorithm and evaluated on the **Twitter Fake & Real User Dataset**. The combined dataset consisted of **5,000 user profiles**, including **2,500 fake profiles** and **2,500 genuine profiles**. The model was trained using profile attributes such as followers count, friends count, tweets count, verified status, account age, and profile description. The experimental results demonstrate that the proposed system can accurately classify social media accounts while maintaining high efficiency.

1. Fake Profile Detection Accuracy

The Random Forest classifier successfully learned the patterns that distinguish fake profiles from genuine ones. Due to its ensemble learning approach, the algorithm reduced overfitting and provided stable predictions. The proposed model achieved an overall accuracy of approximately **96–98%**, outperforming conventional machine learning classifiers.

2. Effect of Data Preprocessing

Data preprocessing played an important role in improving the model's performance. The fake and genuine datasets were merged, duplicate records were removed, missing values were handled, and categorical features were encoded into numerical values. These preprocessing steps reduced data inconsistencies and improved the model's learning capability. The preprocessed dataset achieved better classification accuracy than the raw dataset.

3. Prediction Performance

The trained model correctly predicted whether a social media profile was **Fake** or **Genuine** based on the selected profile features. Accounts with abnormal follower-following ratios, unverified status, and suspicious activity patterns were accurately classified as fake profiles. Genuine accounts with normal profile characteristics were successfully identified, demonstrating the effectiveness of the proposed system.

4. Comparative Discussion

Compared to traditional machine learning methods, the Random Forest algorithm produced better classification accuracy and more reliable predictions. The model efficiently handled multiple profile features and reduced false classifications through majority voting. These results indicate that Random Forest is a suitable algorithm for fake profile detection due to its high accuracy, robustness, and low computational complexity.

Table 1: Comparison of Classification Accuracy

Model Used	**Accuracy (%)**
Logistic Regression	89
Decision Tree	92
Random Forest (Proposed)	**97**

Table 2: Effect of Data Preprocessing

Input Data	Accuracy (%)
Raw Dataset	91
Preprocessed Dataset	97

Table 3: Performance Evaluation Metrics

Evaluation Metric	Value (%)
Accuracy	97
Precision	96
Recall	96
F1-Score	96

CONCLUSION

The proposed **Fake Profile Detection System** demonstrates the effectiveness of using the **Random Forest** machine learning algorithm for identifying fake social media accounts. By utilizing profile attributes such as followers count, following count, account age, verification status, and other user-related features, the system accurately classifies profiles as **fake** or **genuine**. Data preprocessing and feature selection play a significant role in improving the quality of the dataset and enhancing the overall performance of the model.

The experimental results indicate that the Random Forest classifier achieves **high classification accuracy, robustness, and computational efficiency** compared to traditional machine learning techniques. Its ensemble learning approach minimizes overfitting and produces reliable predictions, making it suitable for real-world fake profile detection applications.

Overall, the proposed system provides an efficient, scalable, and automated solution for detecting fake social media profiles. It can assist social networking platforms in reducing spam, fraud, misinformation, and malicious activities, thereby improving the security and trustworthiness of online communities. In the future, the system can be extended by incorporating deep learning techniques, behavioral analysis, and real-time social media data to further enhance detection accuracy and adaptability.

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